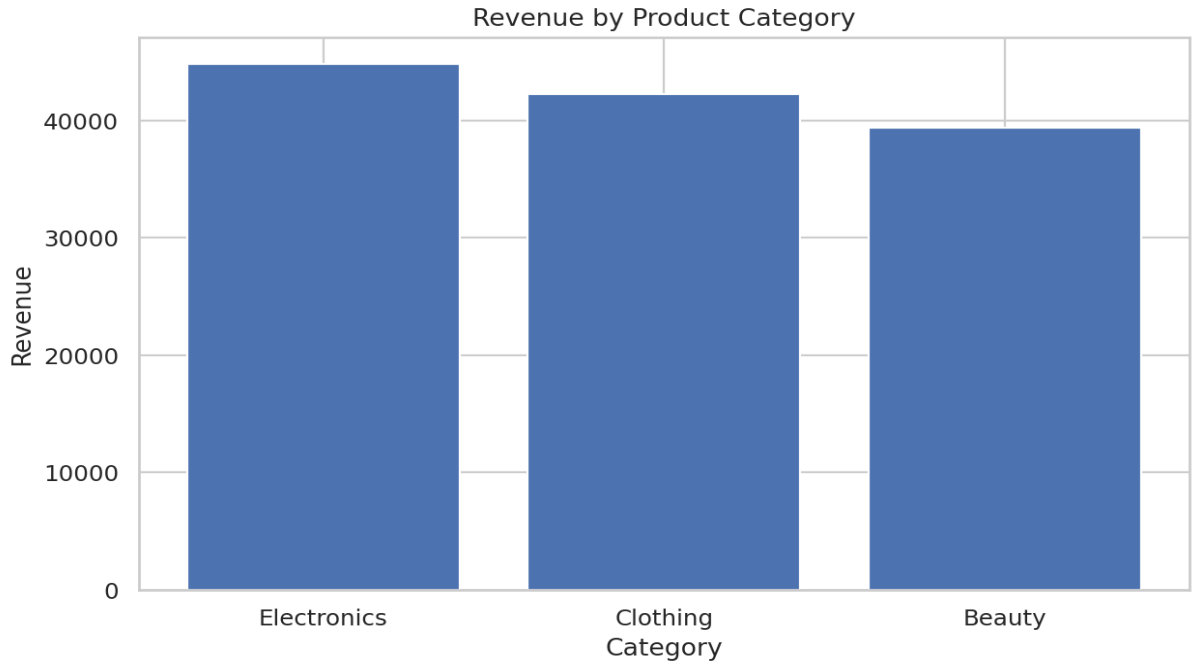
1. Objective

Task 2 extends basic EDA into advanced data wrangling, feature engineering, business analytics and professional reporting.

2. Dataset

The public retail source contains transaction, customer demographic and product-category information. A 251-row working sample is used.

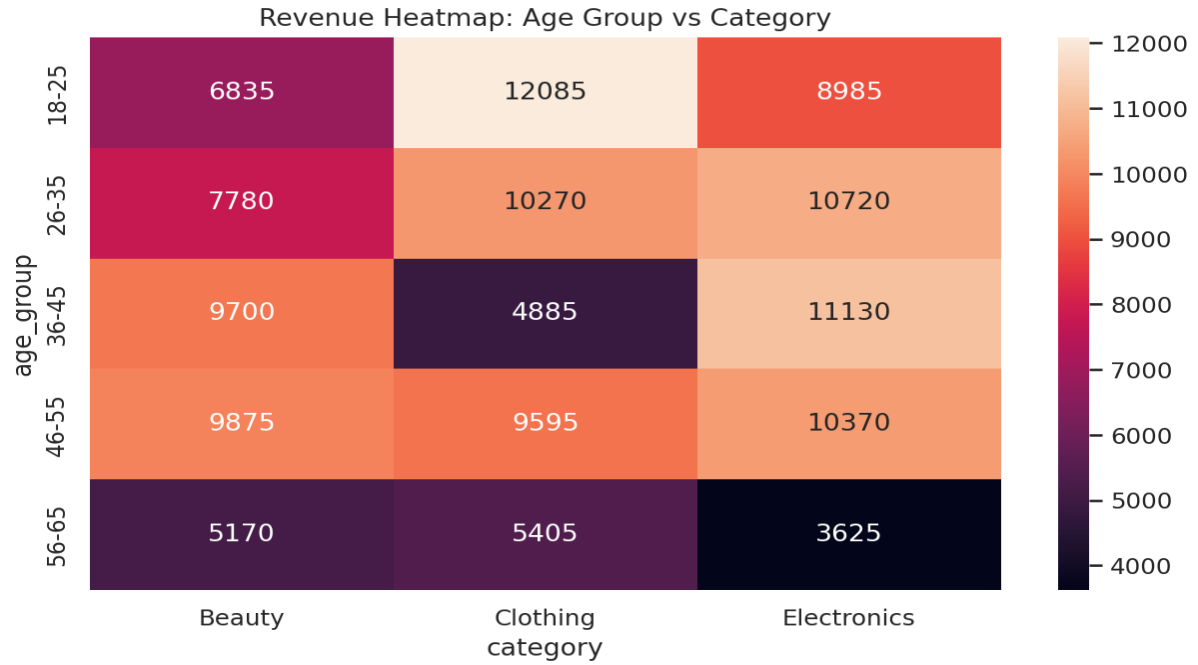


3. Data Integration

The transaction data is normalized into customer, product and sales tables. These are merged with Pandas using explicit keys.

4. Cleaning

The workflow checks data types, duplicates, missing values and arithmetic consistency. The raw source remains unchanged.

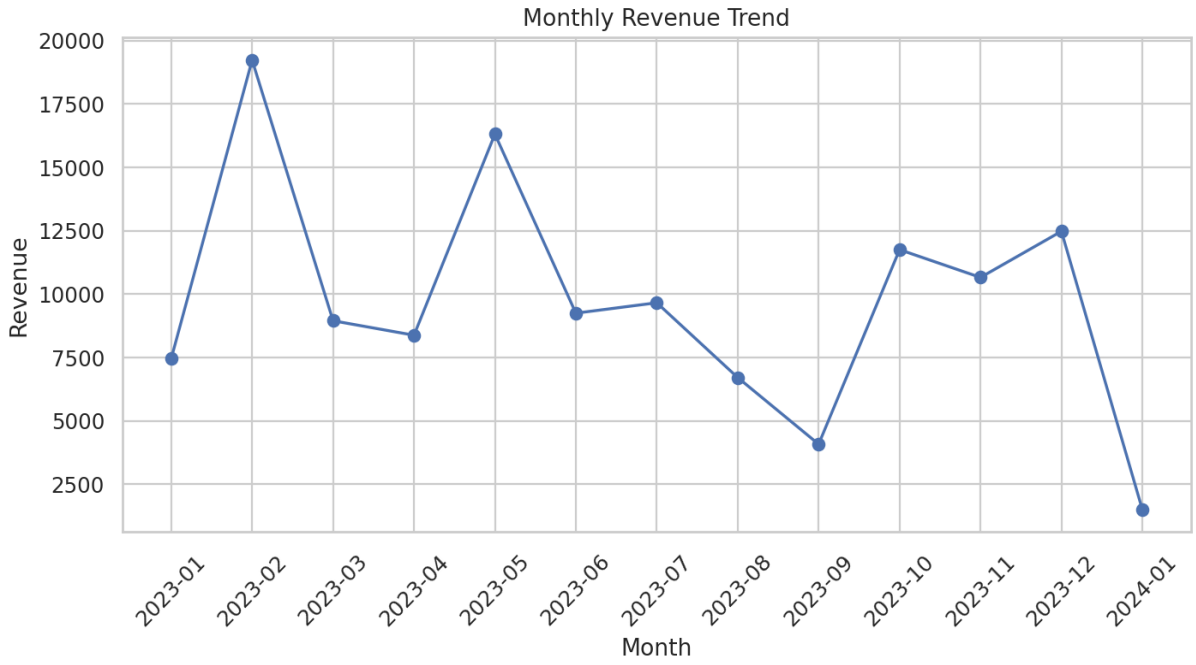


5. Feature Engineering

Month, quarter, year, age group, price band, customer segment and revenue are created as analytical features.

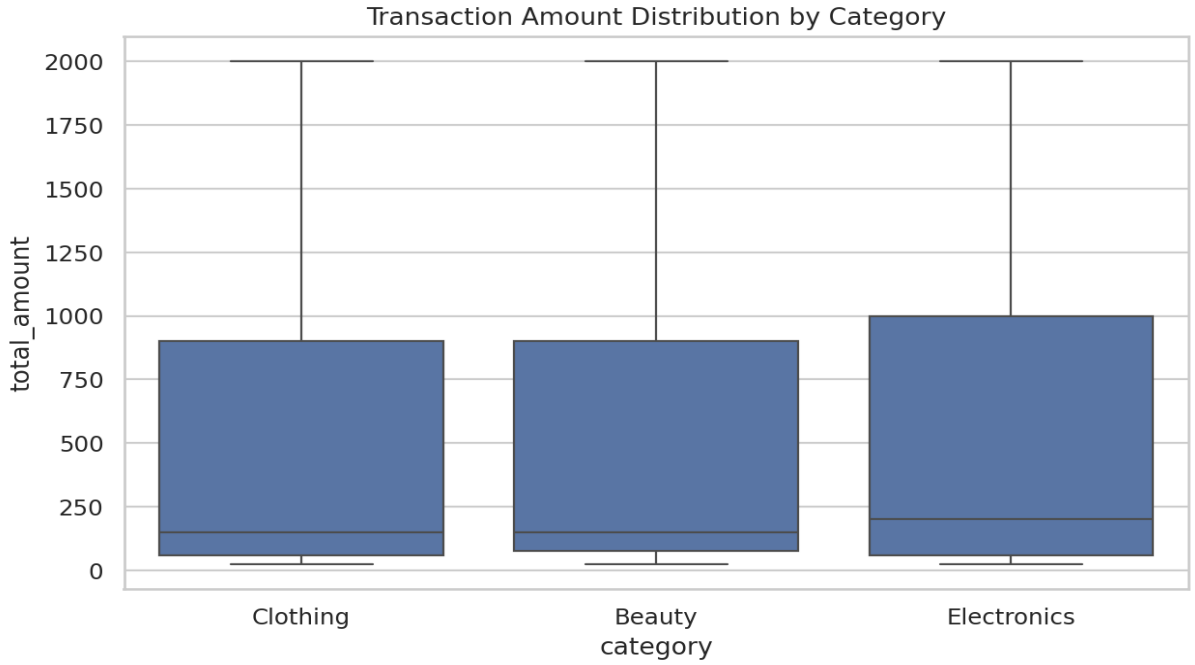
6. Business Analytics

The working sample has 251 transactions, 251 customers and 628 units. Revenue is 126,430; average transaction value is 503.71.



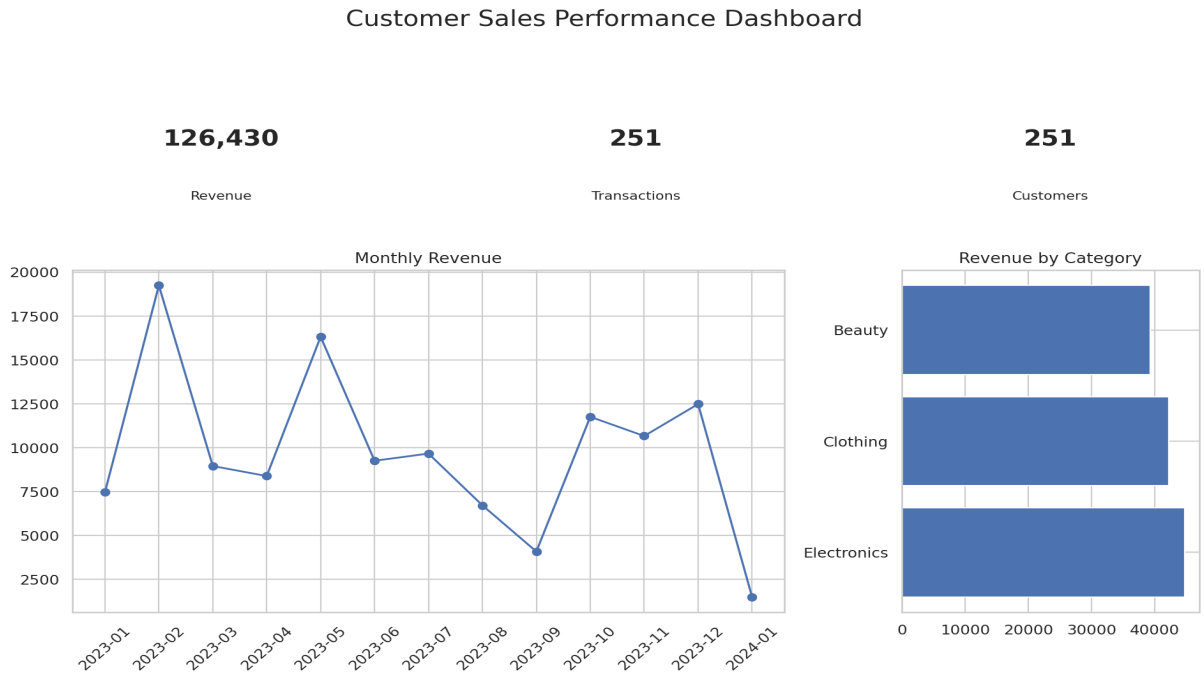
7. Visualization

Matplotlib and Seaborn are used for category, trend, demographic, distribution and relationship charts.



8. Dashboard

The HTML dashboard combines KPI cards, category/gender filters, business charts and a top-customer table.



9. Reporting

The Excel workbook contains multiple business summary sheets and the integrated data. CSV outputs preserve processed tables.

10. Limitations and Conclusion

The source is synthetic and the working sample is smaller than the full file. Product IDs are engineered and profit is not calculated because cost is not available.